

## Math 54 Discussion Worksheet

Name: \_\_\_\_\_ ID: \_\_\_\_\_

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### Problem 1

Let  $\bar{x} = \frac{1}{n}(x_1 + \dots, x_n)$  and  $\bar{y} = \frac{1}{n}(y_1 + \dots + y_n)$ . Show that the least-squares line for the data  $(x_1, y_1), \dots, (x_n, y_n)$  must pass through  $(\bar{x}, \bar{y})$ . That is, if the least-squares line is  $y = \beta_0 + \beta_1 x$ , then we exactly have  $\bar{y} = \beta_0 + \beta_1 \bar{x}$ .

**Problem 2**

Prove the arithmetic mean - geometric mean (AM-GM) inequality:  $\frac{x_1+x_2}{2} \geq \sqrt{x_1x_2}$ . *Hint: Consider the vectors  $u = (\sqrt{a}, \sqrt{b})^T$  and  $v = (\sqrt{b}, \sqrt{a})^T$ .*

**Problem 3**

Consider the space  $C[-1, 1]$  of continuous functions on  $[-1, 1]$  with inner product given by  $\langle f, g \rangle = \int_{-1}^1 \overline{f}g dx$ . Let  $V$  be subspace spanned by  $1, t, t^2$ . Find an orthonormal basis for this space. These are called the Legendre polynomials.